

Platform System Specification

1. Architecture

The platform has four services: the API gateway, the order service, the payment service and the notification service. Services talk to each other over HTTP using JSON. The API gateway is the only component exposed to the internet. All other services run in a private network.

2. Performance targets

The API gateway must answer 95 percent of requests within 300 milliseconds. The payment service must answer 99 percent of charge requests within 2 seconds, excluding the upstream gateway timeout of 10 seconds. Peak load is 500 requests per second. The system must stay available 99.9 percent of every month.

3. Security

All traffic uses TLS 1.2 or higher. Card numbers are never stored; only the gateway token and the last four digits are kept. API tokens expire after 60 minutes. Passwords are hashed with bcrypt. Access to production databases needs two-factor authentication and is logged.

4. Data retention

Order records are kept for 7 years for tax reasons. Application logs are kept for 30 days. Payment audit logs are kept for 5 years. Customer accounts deleted by request are anonymised within 14 days.

5. Disaster recovery

Database backups run every 6 hours and are stored in a second region. The recovery point objective is 6 hours and the recovery time objective is 2 hours. A full recovery drill is done every quarter.

6. Notifications

The notification service sends order confirmation emails within 1 minute of payment success. Failed emails are retried 5 times over 1 hour, then moved to a dead letter queue for manual review.